



Data structure and privacy protection analysis in big data environment based on blockchain technology

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ABSTRACT

In today's digital world, the rapid advancement of Information Technology (IT) has made it crucial to prioritize the protection and management of data storage and retrieval. It is vital to challenge the difficulties related to dispersed and decentralized data to build strong mechanisms for access control and provide effective authorization and authentication in data processing. In the contemporary context of IT, the imperative to secure data storage and retrieval has become distinctly observed. The challenges posed by distributed and decentralized data demand the development of robust mechanisms for access control, demanding a focus on proper authorization and authentication in transaction processing. This research addresses the existing gap by comprehensively adapting data structures effectively to the evolving needs of secure access and storage control. It uses the Enhanced Merkle Tree (EMT) as a novel data structure. This article initially modifies the conventional Merkle Tree (MT) structure used in Blockchain technology to suit e-healthcare Systems (e-HS) requirements. The EMT enhances data security in access and storage and significantly improves data integrity management. Its constant three-degree MT with multiple leaves, branches, and a single root node enables updated data authentication, verification, and validation procedures. The proposed method is applied to the e-HS scenario, and the proposed EMT outperforms existing *state-of-the-art* techniques, achieving a minimal verification time of 14.26 m for 100 transactions. This research, therefore, contributes to the discourse on data security by presenting an innovative and efficient solution tailored to the unique challenges of e-health systems.

1. Introduction

Big Data (BD) has revolutionized data analytics, creating more innovative and intelligent Decision Support Systems (DSS) to discover fascinating facets of such complex and vast data. Although data analytics is valuable to several applications, the flip side is that ever-growing electronic data and its analysis increase alarms about individual privacy attacks. There is large-scale human activity through electronic media, and all these applications collect data in different forms, including personal data. Unauthorized access to these personal data or using collected data for unfair purposes can threaten an individual's privacy. For instance, privacy may be violated when there is access to a person's sensitive data, such as salary, caste, or a disease condition. Analytical techniques can hide sensitive data and concentrate on non-personal data to ensure privacy. However, such techniques focusing exclusively on privacy protection may not give accurate analytical results and fail to improve data quality. Hence, any privacy-aware technology in BD analytics should allow data quality and privacy protection to co-exist. The correctness of a privacy-preserving model depends on

well-defined analytical goals and the context in which the analysis is done [1]. Such a system should ideally balance data quality and privacy protection.

Moreover, the BD scenario must address efficiency and data scale. However, addressing one of these goals reduces the quality of the other. Hence, a scheme that balances these mutually exclusive objectives is a challenge facing privacy-preserving analytics for BD. The Internet and its digital environment have become imperative in all commercial, educational, medical, and industrial applications. Fig. 1 shows the data processing process in the data lifecycle.

Smart contracts optimize the capabilities of Merkle Trees (MT) in Blockchain Technology (BT) by automating the process of verifying data integrity and guaranteeing secure record-keeping that is resistant to tampering. These contracts employ techniques to validate the cryptographic hash values in the MT, ensuring that data remains unchangeable once it is added to the Blockchain. In addition, smart contracts provide conditional execution, the construction of decentralized applications, and the verification of data inclusion or exclusion. Smart contracts are crucial in developing a transparent, reliable, and decentralized system of

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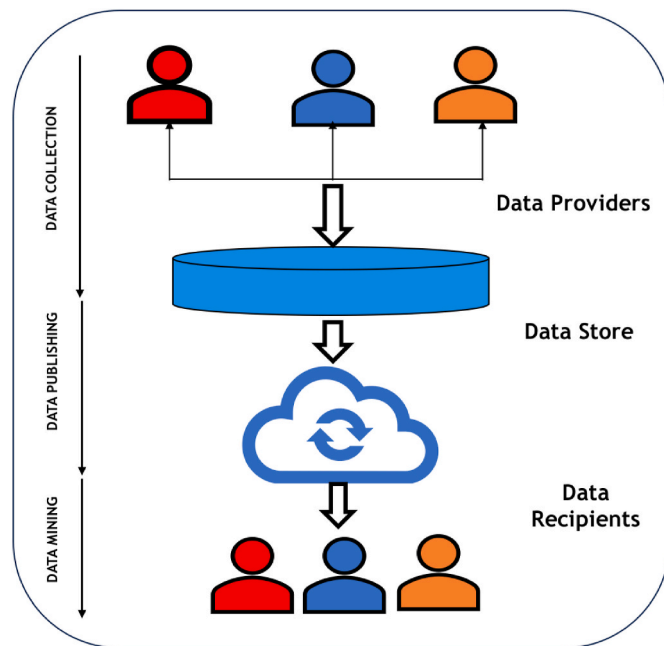


Fig. 1. Data life cycle framework.

programmable capabilities. They enhance security and control by implementing authentication and access restrictions that depend on the verification status of the MT [2].

During the application-level phase, the process involves implementing the investigated blockchain-based data structures and privacy protection measures in actual or simulated BD settings. The primary objective is to smoothly incorporate these solutions into the industry's current systems while targeting specific scenarios or problems typically understood in BD. The practical implementation entails using smart contracts to authenticate users and regulate access, guaranteeing the verification of privacy protection measures. Thorough performance testing evaluates the effectiveness, ability to handle an increased workload, and efficient use of resources of the implemented solutions, focusing on identifying areas that can be improved. The scalability of the BT is assessed to determine its capacity to manage growing quantities of large-scale data. Security assessments are performed to detect and resolve vulnerabilities, while user feedback is collected to evaluate usability and the overall user experience. The application-level phase also guarantees adherence to ethical and regulatory standards, employing a thorough documentation procedure that captures the implementation process, encountered obstacles, and implemented solutions. This phase is crucial for validating the practical feasibility and effectiveness of the investigated solutions in tackling big data difficulties while maintaining privacy protection norms.

Privacy is an all-pervasive concern in today's digital economy. The proliferating use of the Internet and its applications has given rise to a considerable volume of electronic data being collected and processed. Applications that collect these data analyze them to obtain valuable information that can be used for futuristic DSS, increasing the prospects of higher economic value from the data. Hidden patterns in data and unknown relationships between them can improve business value, personalize the user experience, and pave the method toward preferential marketing [3]. Privacy is indeed an issue in worthwhile analytics, and privacy-preserving analytics must ensure that when learning or predicting unusual or otherwise systematic behaviour in data, sensitive or classified information about individuals is not at stake. General Data Privacy Regulations (GDPR) restrict the use of available data, thereby limiting the extent to which collected data can be used, specifically

when used for unintended purposes. Controlling privacy requirements is a challenge since they change concerning a given reproducing scenario, and most often, it is impossible to define a generic privacy rule for all applications. There may be several potential usage scenarios in which a generic privacy rule may diminish the usefulness of data.

Privacy-preserving analytics poses explicit challenges and must adopt approaches that balance size dynamics, data usefulness, performance, and current application requirements. Identifying a suitable balance of these parameters is a challenging search problem involving multiple attributes. These inherent difficulties in privacy-preserving analytics are the driving force behind this research work [4]. Analytics towards research and better DSS is a compelling scenario in today's big data era, and user data privacy is a natural concern. Privacy protection involves applying suitable protection methods to raw data before publication. The data is then said to be privacy-preserved, and the process is generally called anonymization. The methods used for anonymization vary based on application requirements and domain considerations.

The main privacy domains are Statistical Disclosure Control (SDC), Privacy Preserving Data Publishing (PPDP), and Privacy Preserving Data Mining (PPDM). These overlapping domains encompass the study of efficient techniques for privacy protection in data analytics. The caliber of the data a privacy protection system offers end-user applications determines its effectiveness. The goal is to mitigate the risk of sensitive data disclosure while enabling beneficial analysis of privacy-preserved data. However, privacy restrictions inevitably degrade data quality and affect analysis results. Besides, any efficient privacy protection technique for big data should also be able to handle performance and scalability conditions [5].

Incorporating an Enhanced Merkle Tree (EMT) into BT for real-time healthcare applications is an innovative solution that effectively challenges the significant issues of data security, integrity, and accessibility in the healthcare industry. With its streamlined framework for data structuring and cryptographic hashing, the EMT guarantees the immediate integrity of healthcare data. The Blockchain is a decentralized, transparent ledger, securely documenting transactions and changes to the MT while incorporating intelligent contracts that enforce real-time access control regulations, allowing authorized entities to access designated healthcare data. This strategy ensures patient data's confidentiality and safety while establishing an unchangeable record of activities for tracking and responsibility. The integration facilitates compatibility with current Healthcare Systems (HS), providing a smooth layer that improves data protection without interrupting recognized processes. The emphasis is placed on privacy around patients' needs and preferences, giving them complete control over who can access their healthcare data. The decentralized and cryptographically secured system demonstrates resiliency to diverse cyberattacks, ensuring a strong cybersecurity posture. Furthermore, its flexibility enables ongoing enhancements to align with changing healthcare norms and technological progress, rendering it a comprehensive and future-oriented system for managing real-time healthcare data.

With the advances in Information Technology (IT), many healthcare organizations have converted conventional paper-based health records into Electronic Health Records (EHRs). It consists of digitized information on patients' personal data and health records. The digitized health data includes disease details, treatment history, prescriptions, drug summaries, microbiological test reports, and other imaging reports. The considerable number of EHRs and extensive use of healthcare technologies in the big data era increase the role of the Internet and cloud networks to store a large amount of health data and facilitate access across the Network. The insecure nature of Internetwork, third-party maintenance with poor access to data, and financial problems, besides privacy breaches, lead to several attacks [6]. Considering the vulnerable nature of Internetworking, there is a requirement for an immediate and effective method that facilitates the storage, security, sharing, and accessing of health data across numerous stakeholders. Indeed, a secure

patient-centric EHR management system is essential for managing the exponential development of health records. As researchers emphasized, BT promises to revolutionize several industries, including financial transactions, the global supply chain, asset management, healthcare, and decentralized social networking. It stands out as a potent remedy that can fix several flaws in the existing systems.

In this research, the Merkle Tree (MT) data structure reported in BT is enhanced, and a new tree structure that precisely suits the e-health scenario is designed. The main objective of the proposed data structure is to achieve secured storage and integrity management. The proposed data structure is a constant three-degree MT with multiple leaves, branches, and a single Root Node (RN). The proposed EMT is highlighted for data authentication, verification, and validation processes, ensuring integrity. The proposed structure is also applied to the e-Health scenario, and its effectiveness is checked.

1.1. Motivation

The current study field has thoroughly investigated alterations and uses of the MT framework, attempting a wide range of problems in different areas, such as online authentication and the preservation of E-Health record accuracy. Nevertheless, the untapped capacity to mathematically modify the MT while utilizing the SHA-256 cryptographic hash algorithm is a significant deficiency in the existing research model. Although MT has proven value in BT and data integrity verification, incorporating SHA-256 presents a new method to improve performance and security. Inspired by this discrepancy, our study aims to explore the potential collaboration between the MT structure and SHA-256, leveraging the cryptographic capabilities of SHA-256 to enhance data security and integrity. The motivation behind this design initiative stems from the acknowledged constraints of current systems, including compromises in computational efficiency, physical intricacy, and search/verification durations observed in advancements like the Merkle Grid Tree (MG Tree), Distributed Memory Integrity Tree (DMIT), and Adaptive Extended Merkle Tree (AEM-Tree). The recommended update seeks to solve these restrictions by using the recognized attributes of SHA-256. This approach not only tackles present deficiencies but also guarantees compatibility with current systems and is secured against potential risks in the future. The solution integrates mathematically adjusted MT with SHA-256, bridging a research gap and enhancing data security, computational efficiency, and tamper-detection capabilities for critical blockchain applications.

1.2. The research contributions are

- (a) The integration of SHA-256 with the MT structure is being developed to improve data integrity verification in BT applications.
- (b) Addressing limitations in existing systems, the research strives to overcome trade-offs in computational efficiency, complexity, and search times observed in MT variants like MG Tree, DMIT, and AEM-Tree.
- (c) The research significantly strengthens data security by leveraging SHA-256's cryptographic strengths, ensuring robust confidentiality and integrity in critical applications like Blockchain.
- (d) Enhanced computational effectiveness is achieved through optimized performance of the modified MT structure, streamlining authentication, verification, and transaction processing for improved overall system efficiency.
- (e) The MT structure incorporates SHA-256, enhancing its tamper-detection capabilities and, thereby, its ability to detect and respond to unauthorized data modifications.
- (f) The modification addresses current shortcomings and aligns with evolving cryptographic standards, providing a versatile solution for long-term data security needs.

The paper is outlined as follows: Section 2 provides a literature review, Section 3 details the methodology employed, Section 4 is the result and discussion about the proposed work, and Section 6 concludes the study.

2. Related works

The authors [7] explained the MT's capability for online authentication, bolstering resistance against side channels. The Authors have strengthened Secured Embedded Systems by introducing the Frequency Aware Skewed Merkle Tree (FAST) within Field Programmable Gate Arrays (FPGA). By selectively using a skewed MT designed for purposes, FAST reduces the number of traversals. The MG Tree, a mechanism for authorizing and authenticating aggregate queries, was disclosed. They provide support for a range of privacy-preserving queries in their framework.

The study [8] on the Distributed Memory Integrity Tree (DMIT) deviates from the traditional MT to protect memory integrity in distributed shared memory systems and is intended to certify program execution in untrusted environments. The MT is an alternative to public key encryption for device authentication in Smart Grid Networks, a different application. In Ref. [9], we emphasized the significance of tree node optimization for MT's quick and effective production in massive Binary Merkle Tree (BMT) topologies. Furthermore, a Merkle Signature Scheme (MSS) has been designed for quick Hash Tree (HT) generation, considering the requirements of contemporary FPGAs and post-quantum cryptography issues. The MT was expanded to ensure data integrity in conventional geographic data structures. A Distributed Merkle Patricia Tree (DMPT) was designed to improve data authentication in decentralized systems. Although DMPT is less computationally and physically difficult than BMT, block verification may take longer because of DMPT's complex topology [10].

In medicine, In Ref. [11] developed an E-Health Record (HER) integrity system applying a BMT data structure, revealing several issues with managing EHR integrity using traditional BMT. With an emphasis on managing patient profiles with increased privacy, authors [12] presented the Adaptive Extended Merkle Tree (AEM-Tree) data structure for e-Health Security. However, it was discovered that average search, verification, and build times were more extended for AEM trees. In Ref. [13] undertook a thorough examination of immutable data, highlighting the benefits and limitations of each tree structure used to determine content immutability, including Merkle Patricia Trie (MPT), Pattern-Oriented-Split Tree (POS-Tree), and Merkle Bucket Tree (MBT). The MT's data integrity capabilities were combined with a smart contract for user profile verification in Ref. [14] introduction to MT-based authentication for e-learning systems. In Ref. [15], it examined the application of MT in the Wireless Mesh Networks (WMS) domain and developed a safe authentication scheme. A Merkle Tree-based File System (MTFS) was introduced to satisfy private file storage needs. In Ref. [16], a method for Blockchain reliability analysis that uses the Markov Chain Method (MCM) to solve delaying attacks, delay response, and block reliability was presented. Finally, in Ref. [17], the MT-based image authentication approach contributed to image security by assuring image verification and security through BMT.

The research gap in data structures and cryptographic security is the unexplored potential of a mathematically adjusted MT combined with the SHA-256 cryptographic Hash Function (HF). Although MT has been widely used for BT and data integrity verification, there is still a significant knowledge gap regarding how mathematical adjustments, notably the addition of SHA-256, might improve their performance and security [18]. The compelling qualities of SHA-256, which boasts strong security, effectiveness, tamper-resistance, and interoperability with developing cryptographic standards, validate such an effort. This research aims to strengthen data security, boost computational effectiveness, and increase tamper-detection capabilities by combining SHA-256 with the MT structure [19]. It is particularly relevant for

Table 1
Summary of related works.

Ref.	Suggestion	Advantage	Disadvantage
[7]	MT enhances online authentication and strengthens resistance against side channels.	Improved security in online authentication.	It was not specified.
[8]	Introducing FAST, a skewed MT within FPGA, reduces traversals for Secured Embedded Systems.	Increased efficiency through reduced traversals.	Limited information on potential drawbacks
[9]	MG Tree was introduced for authorizing and authenticating aggregate queries supporting privacy-preserving questions.	Supports a range of privacy-preserving queries.	Specific disadvantages are not provided.
[9]	DMIT deviates from traditional MT to protect memory integrity in distributed systems.	Ensures memory integrity and certifies program execution.	Block verification may take longer due to complex topology.
[10]	MT is an alternative to public key encryption for device authentication in Smart Grid Networks.	Provides an alternative for device authentication	No specific disadvantages are mentioned.
[11]	Emphasizes tree node optimization for MT's quick and effective production in massive BMT topologies.	Quick and effective production in massive BMT topologies.	No specific disadvantages are mentioned.
[12]	MSS is designed for quick HT generation, addressing contemporary FPGA and post-quantum cryptography issues.	Quick generation of HT considering FPGA and post-quantum cryptography.	Specific disadvantages are not provided.
[13]	Expansion of MT to ensure data integrity in conventional geographic data structures.	Ensures data integrity in geographic data structures.	No specific disadvantages are mentioned.
[14]	DMPT is designed to improve data authentication in decentralized systems, which is less computationally and physically difficult than BMT.	Improves data authentication in decentralized systems.	Block verification may take longer due to complex topology.
[15]	The e-HER integrity system utilizes the BMT structure, revealing EHR integrity issues.	Identifies issues with managing EHR integrity using traditional BMT.	No specific advantages are mentioned.
[16]	AEM-Tree presented for e-Health Security, with extended search, verification, and build times.	Enhanced security for e-HS, but with extended search, verification, and build times.	Extended search and processing times are noted drawbacks.
[17]	A thorough examination of immutable data and tree structures, highlighting benefits and limitations	Provides insights into the benefits and limitations of several tree structures.	No specific advantages or disadvantages are listed.
[18]	MT-based authentication for e-learning systems using smart contracts for user profile verification	Utilizes MT for secure authentication in e-learning systems.	Specific advantages and disadvantages were not detailed.
[19]	Examines the application of MT in Wireless Mesh Networks (WSN), developing a safe authentication scheme.	Develop a secure authentication scheme for WMN.	Specific advantages and disadvantages were not detailed.
[20]	Presents a method for Blockchain reliability analysis using the MCM.	It uses the MCM to address delaying attacks and block reliability.	No specific disadvantages are mentioned.
[21]	MT-based image authentication approach contributes to image security through BMT.	Assures image verification and protection through MT-based image authentication.	No specific disadvantages are mentioned.
[22]	Introduces the unexplored potential of a mathematically adjusted MT combined with SHA-256.	It aims to strengthen data security and increase tamper-detection capabilities.	No specific disadvantages are mentioned.
[23]	Addresses the knowledge gap on adding SHA-256 to MT for improved performance and security.	It aims to improve computational effectiveness and interoperability.	No specific disadvantages are mentioned.
[24]	Summarizes research contributions covering diverse fields, from healthcare to climate change, using BT.	Blockchain applications are showcased in diverse fields, emphasizing benefits and potential drawbacks.	Specific advantages and disadvantages for each application are not detailed.

applications like BT, where trust and data integrity are crucial. This method also improves interoperability with current systems and gives a mechanism for future-proof against increasing security threats, filling a critical research gap [20].

The compilation of research papers covers several fields (Table 1), attempting significant problems in healthcare, agriculture, climate change, big data, and cloud computing by employing inventive uses of BT. The initial study presents a method to protect privacy in e-HS based on the Internet of Things (IoT) [21]. This method utilizes BT to create and manage health documents, guaranteeing security by implementing intelligent contract regulations. Incorporating Blockchain, cryptography, and IPFS technologies in agriculture provides a detailed data access control system, ensuring the security and confidentiality of data in agricultural IoT. The third study examines the convergence of BT and climate change, specifically focusing on its potential applications in carbon trading and sustainable energy [22]. It emphasizes the advantages of Blockchain in terms of transparency, traceability, and immutability while also addressing Energy Consumption (EC) problems. In big data, the Data Trust Method is implemented to improve the accuracy of predictions for single-variable time series [23]. It achieves this by effectively detecting and replacing unreliable data throughout forecasting. Finally, the study on cloud and Edge computing presents a decentralized architecture based on Blockchain, showcasing enhanced system availability, even in situations where the network is divided, without sacrificing configuration integrity [24].

3. Proposed methodology for enhanced Merkle Tree (EMT)

The proposed EMT data structure, intended to guarantee data security for crucial data-driven applications, is given in this subsection in detail. The four main functional elements that comprise the system's architecture are Transactions, a 3-stage EMT, Block Elements, and a Chain of Blocks. Together, these functional parts perform the following four vital tasks: generation of EMT, block generation, data synchronization, and data authentication. The architecture of the proposed EMT data Structure for integrity management is presented in Fig. 2. It depicts the process of hash tree generation with the EMT data structure with sample input transactions.

The method of recovering data in an MT system based on BT is carefully and complexly handled by utilizing the inherent characteristics of BT and the structure of the MT itself. The inherent unchangeability of Blockchain guarantees that once data is incorporated into a block and appended to the chain, it stays impervious to unauthorized alterations and maintains its security [25,26].

Periodic backups of the complete Blockchain, dispersed among nodes in the decentralized network, propose a safeguard in the event of data loss. The MT's architecture, with cryptographic hashes connecting data blocks, enables rapid verification of integrity and rebuilding in the event of unintentional loss. Smart contracts can automate the recovery procedures activated by predetermined criteria. Timestamps assigned to each block allow for the systematic recording of chronological order, making it easier to pinpoint precise moments in time for data retrieval. Collaboration between nodes in the network guarantees that in the event

of data loss in one node, it can synchronize with other nodes to recover the missing information. Ensuring the secure administration of private keys is crucial to prevent unauthorized access and uphold the system's overall resilience. These techniques work together to establish a strong framework for retrieving data, focusing on preventing issues in advance and reconstructing data in response within MT topologies based on BT.

Data protection legislation, such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), protect individual privacy and promote responsible data handling. The General Data Protection Regulation (GDPR), enforceable throughout the European Union and European Economic Area, sets forth fundamental values such as lawfulness, fairness, and transparency. It emphasizes limiting the purposes for which data is collected, minimizing the data collected, ensuring its accuracy, and holding organizations accountable for their data processing activities. The legislation bestows individuals with valid personal data and enforces substantial penalties for failure to comply. Conversely, HIPAA, a

chained with other blocks and form a Blockchain. A detailed explanation of the input, output, and process involved in each functional component of the proposed architecture is explained in the subsequent subsections.

3.1. Enhanced Merkle Tree construction (EMTC)

The EMTC construction starts with collecting data components or transactions that need to be secured and organized before creating an EMTC. A HF is applied to each data element to make Leaf Nodes (LN) for the tree (e.g., *SHA-256*). These LN represent the various data elements. The next level of the tree is formed by combining pairs of LN to produce intermediate nodes (*hashed*). Hashing and combining nodes in this manner will continue until you reach a single RN. This RN represents the complete data collection.

Algorithm 1. SHA-256

Step 1. Initialize constants and hash value H (initials)

Step 2. For Each block M_i in the padded message:

Step 3. Prepare message schedule ' W ' from ' M_i ' and the previous schedule.

Step 4. Initialize working variables:

Step 5. $a=H[0]$; $b=H[1]$; $c=H[2]$; $d=H[3]$; $e=H[4]$; $f=H[5]$; $g=H[6]$; $h=H[7]$;

Step 6. Perform 64 rounds of processing:

$$\mathbf{T}_1 = \mathbf{h} + \sum_1 (\mathbf{e}) + \mathbf{Ch}(\mathbf{e}, \mathbf{f}, \mathbf{g}) + \mathbf{K}[\mathbf{t}] + \mathbf{W}[\mathbf{t}]$$

Step 7.

$$\mathbf{T}_2 = \sum_0 (\mathbf{a}) + \mathbf{M}_{aj}(\mathbf{a}, \mathbf{b}, \mathbf{c})$$

Step 8.

Step 9. $h=g$; $g=f$; $f=e$; $e=d + T1$; $d = c$; $c = b$; $b = a$; $a = T1 + T2$;

Step 10. Update H :

Step 11. $H[0] += a$; $H[1] += b$; $H[2] += c$; $H[3] += d$; $H[4] += e$; $H[5] += f$; $H[6] += g$; $H[7] += h$;

Step 12. The final H is the SHA-256 hash of the input message.

federal statute in the United States of America (USA), specifically aims to safeguard health information by implementing the Privacy, Security, and Breach Notification Rule. HIPAA provides individuals the right to access their health information, requires measures to protect e-HS data, and imposes fines for any breaches. Both legislation highlights the worldwide trend towards strong data protection, as corporations implement strict measures to ensure transparency, security, and the preservation of individuals' rights regarding personal and sensitive information.

The functional transaction part of the proposed architecture aims to take care of input data to be fed to the Blockchain. The second functional component of the proposed III-stage EMT is the new data structure that can efficiently take care of the security of the input data with the hashing process. The novelty of the proposed III-stage EMT is not only in its design construction but also in achieving security besides helping integrity management. The proposed architecture creates blocks, consisting of a few elements called block elements, forming the proposed architecture's third functional component. The block elements include Block Number, Nonce, Timestamp, Previous Hash, Block Data, and Block Hash. These collective elements constituting a Block ' B_i ' are

A prominent cryptographic hash algorithm that creates a fixed-size (256-bit or 32-byte) hash result from an input message is called SHA-256 (Secure Hash Algorithm 256-bit). A mathematical model is comprised of the *SHA-256* algorithm that is provided in this section. Let's denote the following: ' M ' stands for the input message, which is made up of a series of bits; ' H ' for the hash computation state, which is initially set to a fixed value known as the initial hash value, ' K ' for a set of constants used in the algorithm, and ' W ' for an array of 64 words, each of which contains 32 bits for calculations and temporary storage. $F_i(X, Y, Z)$ indicates one of the logical functions applied in *SHA-256*. $\Sigma_0(X)$ and $\Sigma_1(X)$ indicate bitwise rotation operations. $\mathbf{Ch}(X, Y, Z)$ and $\mathbf{Maj}(X, Y, Z)$ indicate logical functions applied in *SHA-256*. The *SHA-256* algorithm can be detailed in several steps:

Add a padding to the message ' M ' to ensure its length is a multiple of 512 bits (64 bytes). Append the length of the original message in bits as a 64-bit representation. Divide the padded message into 512-bit blocks, denoted as $1, 2 \dots, M_1, M_2 \dots, M_n$. Break each M_i into 16-bit and 32-bit words to create the message schedule ' W '. The hash value H to the initial hash values is initialized. For every block M_i , prepare the message schedule W using the current block M_i and the previous message

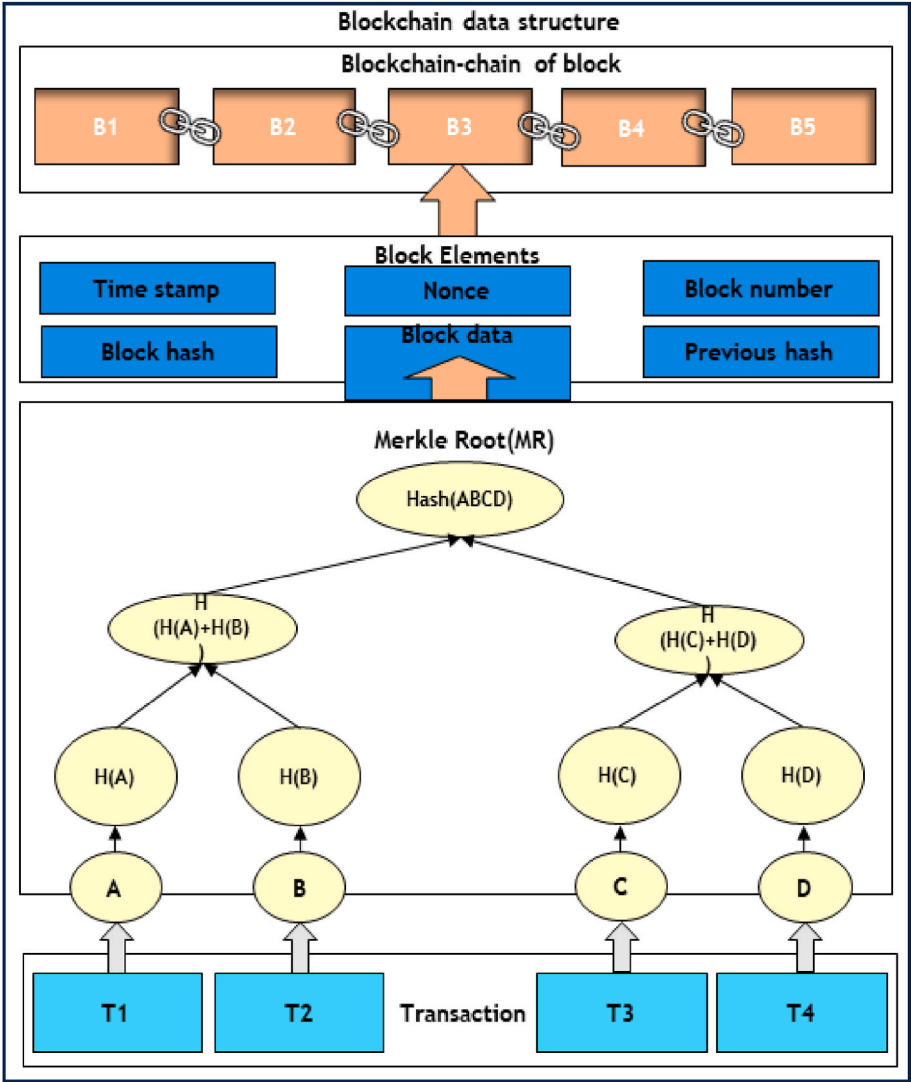


Fig. 2. Architecture of EMT.

schedule. Perform 64 rounds of processing using a loop and update 'H' in each round using EQU (1) and EQU (2). After processing all blocks, the final hash value 'H' is the SHA-256 hash of the input message 'M'. The SHA-256 procedure is given in Algorithm 1.

$$T_1 = h + \sum_1 (e) + Ch(e, f, g) + K[t] + W[t] \quad (1)$$

$$T_2 = \sum_0 (a) + M_{aj}(a, b, c) \quad (2)$$

The SHA-256 algorithm includes additional logical functions, bitwise operations, and constants to ensure cryptographic security. Let 'D' represent the data elements or transactions to be organized in the EMT. The $H(x)$ is the HF determined and utilized to calculate the hash of a data element 'x'. L_i represents the LN, where $L_i = H(D_i)$ for each data element D_i in 'D'. ' I_j ' represents the intermediate nodes, where $I_j = H(I_{2j-1} || I_{2j})$ is formed by concatenating and hashing the child's nodes I_{2j-1} and I_{2j} . R represents the RN, the final hash value $R = H(I_1)$.

LN calculation is accomplished using EQU (3)

$$L_i = H(D_i) \quad (3)$$

Intermediate Node calculation is accomplished using EQU (4)

$$I_j = H(I_{2j-1} || I_{2j}) \quad (4)$$

RN calculation is accomplished using EQU (5)

$$R = h(I_1) \quad (5)$$

3.1.1. Code for data deletion in tree object

To enable data deletion in the system, garbage collector services manage to delete orders from the data authenticator.

```

{
  Header
  {
    Generate ID || Date & Time || Schema_ID || Input_Src.
    {
      ...;
    }
    User_ID || Test_ID;
    {
      Update Features
      {
        Seq_ID;
        Delete = 0;
      }
    }
  }
  Body
}

```

3.1.2. Code for data deletion in tree LN

As datasets develop, methodologies to reduce tree size will become mandatory.

```

{
  Header
  {
    Generate ID || Update Features
    {
      Seq_ID;
    }
  }
}

```

3.1.3. Code for tree traversal

The tree is traversed recursively in preorder. Here's an example of the traversal algorithm.

```

Node = RN ( )
Tree Traverse (Node )
If
{
  Hit Rate;
}
Break;
If
{
  Miss Rate;
}
If
{
  Node == LN;
}
Break;
Tree Traverse
{
  Node.Left;
  Node.Right;
}

```

3.2. Block creation

When using a blockchain or any other data structure that uses an EMT, creating a block requires several steps. The RN of the EMT acts as a summary of the data contained within the block. The block header then includes this RN, often a cryptographic hash. The block is then filled with relevant transactions or data pieces. The entire content, including the header and contents, is put through a hashing procedure to ensure data integrity and provide a unique identifier for the block, producing a recognizable hash value. The previous block's hash is finally linked to the current one, producing an interlinked chain of blocks to create the chronological order and continuity of the Blockchain.

Table 2Comparison of *BMT* with *EMT* for eight input transactions.

Transactions	EMT			BMT		
	Verification Node	BT	Higher Order Nodes	Verification Node	BT	Higher Order Nodes
T ₁	H ₁₂ , H ₁₁ , H ₁₀ , H ₂	4	3	H ₁₄ , H ₁₀ , H ₂	6	1
T ₂	H ₁₂ , H ₁₁ , H ₁₀ , H ₂	4	3	H ₁₄ , H ₁₀ , H ₁₁	6	1
T ₃	H ₁₂ , H ₁₁ , H ₉ , H ₄	4	3	H ₁₄ , H ₉ , H ₄	6	1
T ₄	H ₁₂ , H ₁₁ , H ₉ , H ₃	4	3	H ₁₄ , H ₉ , H ₃	6	1
T ₅	H ₁₂ , H ₁₀ , H ₉ , H ₆	4	3	H ₁₃ , H ₁₂ , H ₆	6	1
T ₆	H ₁₂ , H ₁₀ , H ₉ , H ₅	4	3	H ₁₃ , H ₁₂ , H ₅	6	1
T ₇	H ₁₁ , H ₁₀ , H ₉ , H ₈	4	3	H ₁₃ , H ₁₁ , H ₈	6	1
T ₈	H ₁₁ , H ₁₀ , H ₉ , H ₇	4	3	H ₁₃ , H ₁₁ , H ₇	6	1

Let 'B' be a block, $B=(H_{prev}, D, HB)$, where H_{prev} is the previous block's header hash. D is the data or transactions to be included in the block. 'HB' is the hash of the entire block header and data, i.e., $HB=H(B)$.

Block Hash Calculation is accomplished using EQU (6).

$$HB = H(H_{prev} || D) \quad (6)$$

3.3. Data authentication

The inherent characteristics of MT are used to authenticate data within an *EMT*. One can take several actions to maintain the integrity of a specific data element within a block. The *EMT*'s root hash is acquired for that initial block. Further, request the hashes of the sibling nodes located along the path that connects the LN, which symbolizes the significance of the data element and the tree's root. This path can be rebuilt by repeatedly performing a hashing function to the data and including its sister hashes until the root hash is achieved. Finally, check the recalculated root hash with the one provided to ensure the accuracy and integrity of the data. The data is still authentic and unchanged if the hashes of the two files match.

Let B be a block, $B=(H_{prev}, D, HB)$, where H_{prev} is the previous block's header hash. D is the data or transactions to be included in the block. HB is the hash of the entire block header and data (i.e.) $HB=H(B)$.

Authentication path scheming is accomplished using EQU (7).

$$P = (H_i, S_1, S_2, \dots, S_k) \quad (7)$$

Authentication check is accomplished using EQU (8).

$$H_i = H(S_1 || S_2 || \dots || S_k || R) \quad (8)$$

Within an *EMT* coupled with a BT, the management of cryptographic keys encompasses multiple entities to guarantee the secure handling of these keys. The key generator is tasked with producing keys utilized in the *EMT* framework, which is essential for upholding data integrity. A Key Distribution Center (KDC), whether centralized or distributed, can assist in the secure distribution of session keys throughout the BT. The Public Key Infrastructure (PKI) encompasses many components, such as Certificate Authorities (CAs) and Registration Authorities (RAs), which are responsible for the management of digital keys and certificates linked with the *EMT*. Users or apps interacting with the *EMT* within the Blockchain depend on key custodians, Hardware Security Modules (HSMs), and maybe decentralized key servers to store and distribute keys securely. Ensuring the integrity and security of the *EMT* data within the BT requires the vital coordination of these entities.

3.4. Data synchronization

Data synchronization centres around maintaining data reliability across many copies inside a distributed system in the context of an *EMT* (Table 2). Updates are made to the impacted LN in the corresponding *EMT* when changes are made to one data instance. The hashes along the way to the root are simultaneously updated and adjusted to consider these changes. The Network's relevant nodes and locations are then informed of this updated root hash. By matching their root hashes to the most recent propagated root hash, nodes in this cooperative network design verify data synchronization. A matching result means the data is synchronized and coordinated throughout the distributed system.

Data synchronization involves updating an LN and propagating the changes to the root and across the network. Let D_{new} be the new data for a LN. The synchronization is done using EQU (9).

$$H_{new} = H(D_{new}) \quad (9)$$

The LN is updated using EQU (10).

$$L_i = H_{new} \quad (10)$$

Recalculate the intermediate and RNs as necessary, updating I_j and R . New Leaf Node Calculation is accomplished using EQU (11).

$$L_i = H(D_{new}) \quad (11)$$

Update Intermediate and RN, as necessary. Recalculate the intermediate nodes and RN using the same formulas used during *EMT* Generation, EQU (12).

$$\text{Enhanced Merkle Tree (EMT) + Binary based Merkle Tree (BMT)} \quad (12)$$

In this subsection, the process of constructing an *EMT* data structure with increased transactions, say eight input transactions for the same healthcare application, is presented.

Let us consider the situation of patient P_1 consulting a specialty doctor, 'D₂', for a disease at hospital H_2 . In the consultation process of P_1 , the hospital H_2 ought to have access to P_1 's entire previous health history. With P_1 success permission, hospital H_2 retrieves the previous health records of P_1 . While retrieving the health records, the proposed system verifies the integrity of stored transactions.

Let us now consider a situation where P_1 has the following eight transactions: t_1 =Prescription, t_2 =Drug Summary, t_3 =Microbiological Test Report-1, t_4 =Microbiological Test Report-2, t_5 =Ultra Sonic Scan Report-1, t_6 =Ultra Sonic Scan Report t2, t_7 =Operation Summary and t_8 =Bill Payment at hospital H_2 . With these identified transactions, the proposed *EMT* data structure generates the leaf hashes for all input transactions. Then, the proposed system concatenates the generated hashes and forms the branch nodes for the eight input transactions. Finally, the proposed system generates one Merkle Root, as described in the previous section. A new block B2 is created for P1 at hospital H2 to store all the transaction hashes. The *EMT* construction requires 13 nodes to generate the MT for the eight input transactions. The *EMT* data structure with 8 input transactions for the healthcare scenario is presented in Fig. 3.

For comparison purposes, the construction of the existing *BMT* with eight input transactions is presented in Fig. 4. It is clear from Fig. 4 that the existing *BMT* requires as many as 15 nodes for the construction of the tree, in contrast to 13 nodes in the proposed *EMT* construction. As the number of transactions increases, the *BMT* grows exponentially, but in the case of the proposed *EMT*, the degree is kept constant.

Table 1 compares the *BMT* and the *EMT* regarding the number of verification nodes, binary trees, and higher-order nodes. The *BMT* has 6 trees and only one higher-order node for each transaction verification procedure, compared to the proposed system's four binary trees and three higher-order nodes.

The hashing process increases when trees are built with fewer higher-order nodes because more inter-level branch nodes are needed. The resulting tree has a narrower span and more nodes than a MT. The

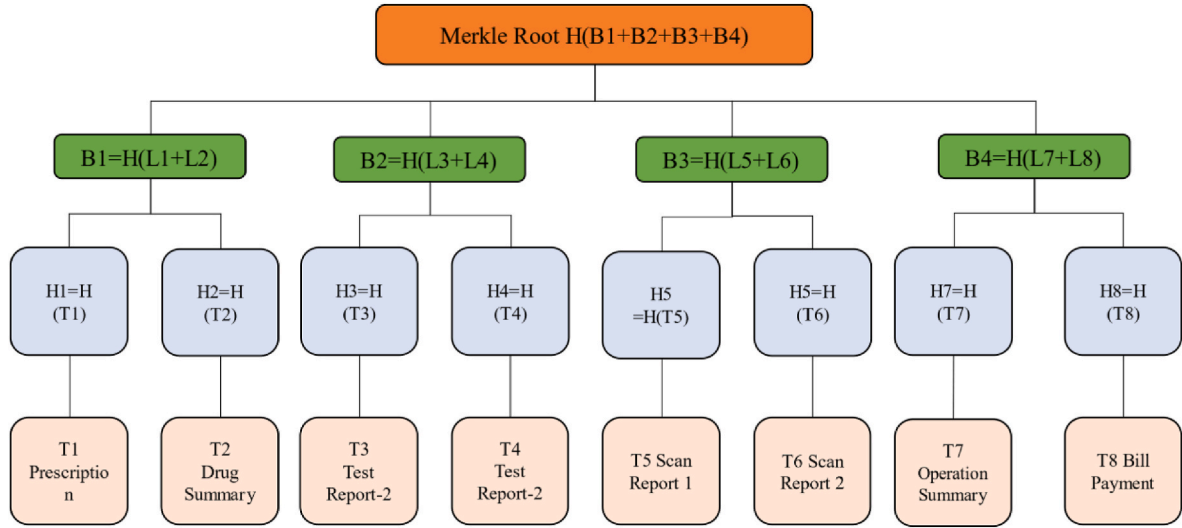


Fig. 3. Construction of *EMT* for healthcare application with eight input transactions.

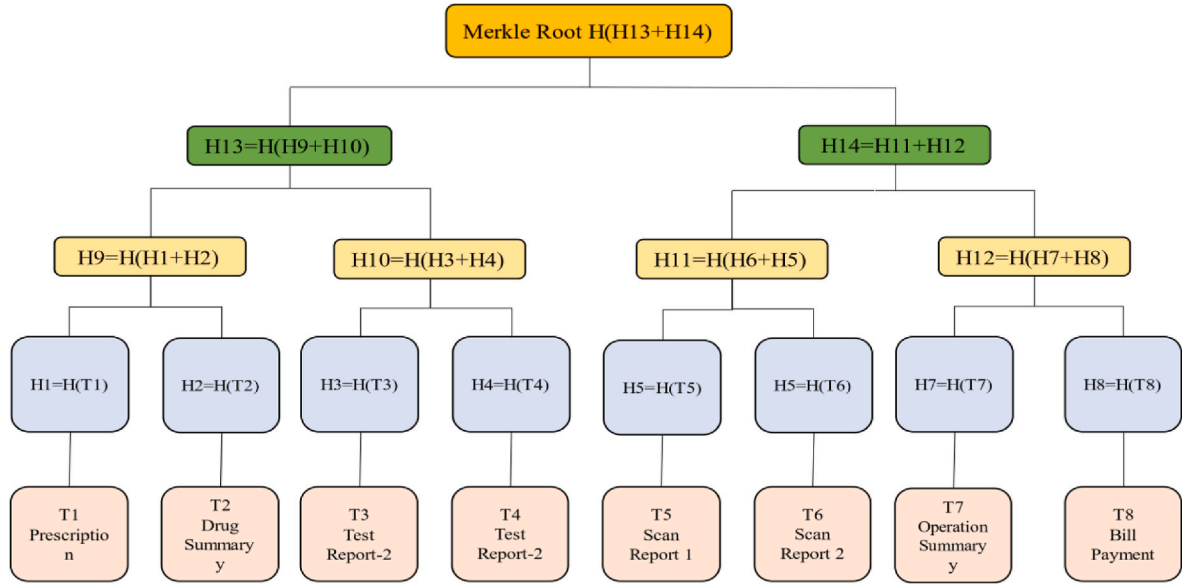


Fig. 4. Construction of *BMT* for healthcare application with eight input transactions.

total number of verification nodes, the total number of BT contained within the tree structure, and the total number of higher-order nodes inside the tree structure are generally used to measure the system's overall complexity. These higher-order nodes follow the RN or Merkle Root in order. It is noted that the MT's construction is noticeably more complicated than the *EMT*s when dealing with eight input transactions in a traditional BMT.

The computational complexity of the proposed system is also evaluated based on the number of verification nodes, particularly those that are higher-order nodes situated closest to the Merkle Root. In situations where a MT contains a more significant number of lower-order nodes, even if the overall count of verification nodes is lower, the process of hash path verification becomes more time-consuming. The presence of numerous lower-order nodes generates numerous BT within the MT, consequently increasing the height of the MT. As the tree becomes more densely branched, identifying the hashes within the MT becomes significantly more complex, in addition to the increased storage requirements. However, the proposed *EMT* has been demonstrated to have fewer inner branches, mitigating these complexities.

The novelty of the proposed *EMT* construction increases the number of branches at the root level, where the branches are attached to binary leaves instead of growing new branches. Also, it can be observed that the proposed *EMT* contains a minimum of one leaf and a maximum of two leaves at the lower level, and hence, the time for constructing the *EMT* shall be significantly reduced compared to existing Merkle Trees. The performance of the proposed construction of *EMT*, Validation, and Synchronization processes are experimented with by implementing the BT, and the same is presented in the next section with the results obtained.

4. Result and discussion

4.1. Experimental environment

This work implemented and experimented with the proposed system to evaluate its efficacy on a Windows 10 computer with a 1 GHz Intel Core i3 processor and 4 GB of memory. Python version 3.8.4 was used throughout the proposed system's implementation. The research is

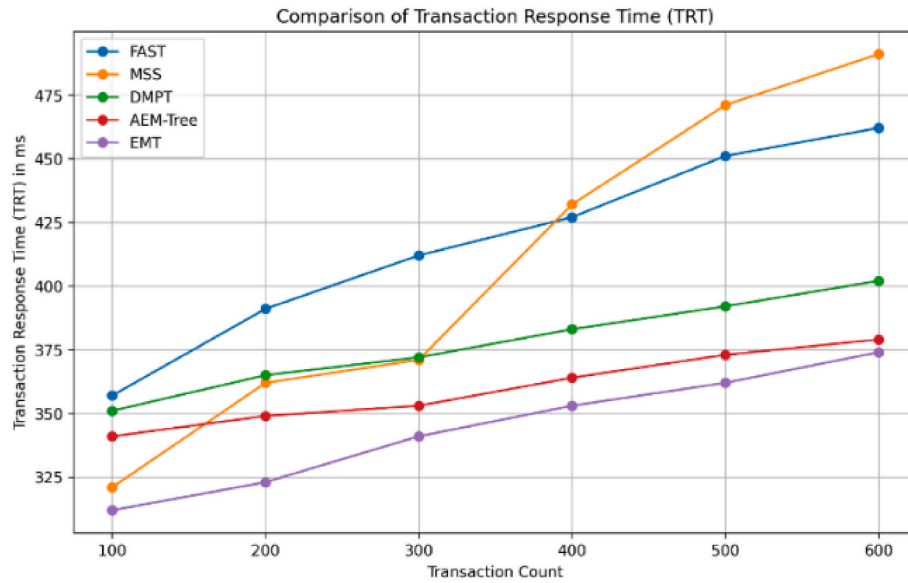


Fig. 5. Comparison of TRT.

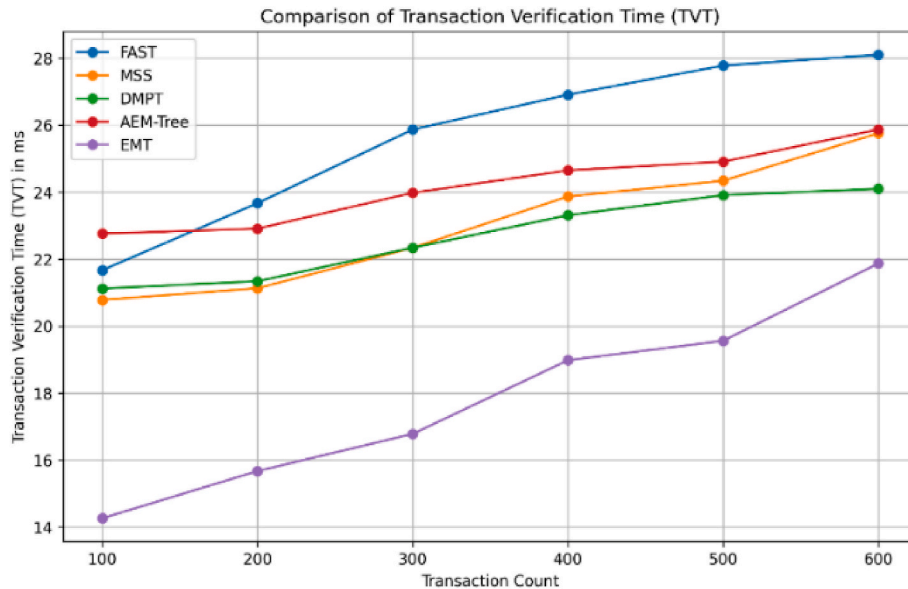


Fig. 6. Comparison of TVT

completed on several tasks using the Python IDLE environment, including creating the *EMT* with hashed transactions, creating blocks, hashing blocks, creating chains of blocks, confirming the existence of input data, and validating block reliability. The research uses real-time medical records of 2 GB. The performance of the proposed *EMT* data structure has been evaluated by conducting laborious experiments, which are presented in this section. The proposed *EMT* is compared with the existing techniques, namely *FAST*, *MSS*, *DMPT*, and *AEM-Tree* (Figs. 5–8), which utilizes a dataset.

For implementing an *EMT* for medical data, Hyperledger Fabric stands out. As a permission blockchain, it prioritizes data security and privacy. Its modular architecture and support for smart contracts enable the creation of private networks with fine-grained access controls. Hyperledger Fabric's suitability for healthcare lies in its capacity to ensure the integrity and confidentiality of medical records, aligning with regulatory requirements. Collaborating with BT and healthcare compliance experts is crucial for designing and implementing a secure

system that addresses the specific challenges of managing sensitive medical information within a distributed network.

4.2. Performance evaluation

To validate the effectiveness of the proposed system, standard measures viz Transaction Response Time (*TRT*) (Table 3), Transaction Verification Time (*TVT*) (Table 4), Throughput (Table 5), and Delay/Latency (Table 6) are used. The metric *TRT* is the time required to complete the encryption process for the input transactions on a particular block. The second metric in the experimental design is delay. It measures the delay between requests. It is an essential metric for identifying delays in *TRT*. The system's performance refers to the maximum rate at which the system processes requests. Through this latency metric, the system can assess its feasibility and scalability. If the delay increases, it can negatively impact the system's performance. The time needed to verify and approve a transaction on the Blockchain is known as *TVT*. It

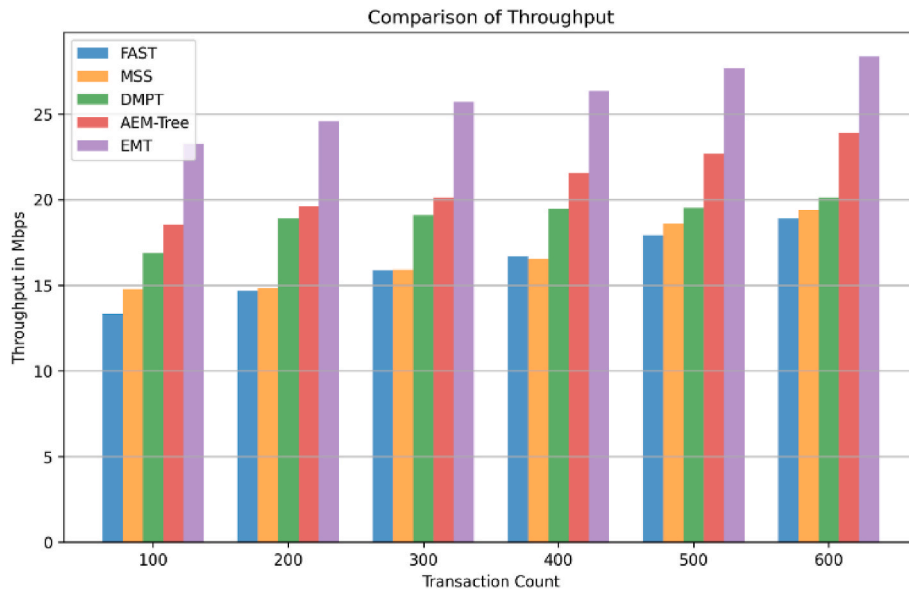


Fig. 7. Comparison of throughput.

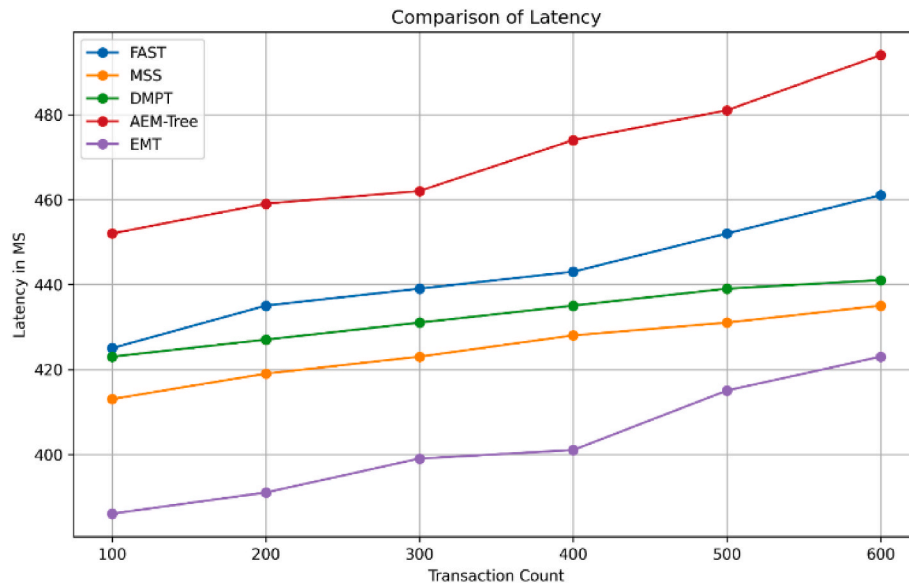


Fig. 8. Comparison of latency.

Table 3
Comparison of TRT.

Algorithm	100	200	300	400	500	600
<i>FAST</i>	357	391	412	427	451	462
<i>MSS</i>	321	362	371	432	471	491
<i>DMPT</i>	351	365	372	383	392	402
<i>AEM-Tree</i>	341	349	353	364	373	379
<i>EMT</i>	312	323	341	353	362	374

Table 4
Comparison of TVT.

Algorithm	100	200	300	400	500	600
<i>FAST</i>	21.67	23.67	25.87	26.91	27.78	28.1
<i>MSS</i>	20.78	21.13	22.34	23.87	24.34	25.76
<i>DMPT</i>	21.12	21.34	22.34	23.31	23.91	24.1
<i>AEM-Tree</i>	22.76	22.91	23.98	24.65	24.91	25.87
<i>EMT</i>	14.26	15.67	16.78	18.98	19.56	21.87

begins when a user initiates a transaction and concludes when it has been successfully validated, added to a block, and has become an irreversible blockchain component. Lower TVT is preferred as it indicates quicker transaction processing and verifies that the BT can effectively handle many transactions. In the context of Blockchain, throughput refers to how many transactions a network can process and confirm in a specific amount of time, typically measured in Transactions Per Second (TPS). A BT with a higher throughput is more scalable, efficient, and able

to process a larger volume of transactions, making it suited for applications with high transaction demands, such as financial systems or supply chain management.

4.3. Discussion

The analysis of TRT for different algorithms shows that FAST reliability increases TRT as Transaction Count increases, while MSS shows

Table 5
Comparison of throughput.

Algorithm	100	200	300	400	500	600
<i>FAST</i>	13.34	14.67	15.87	16.68	17.9	18.9
<i>MSS</i>	14.78	14.81	15.91	16.56	18.6	19.4
<i>DMPT</i>	16.87	18.89	19.1	19.45	19.51	20.12
<i>AEM-Tree</i>	18.56	19.63	20.12	21.56	22.67	23.89
<i>EMT</i>	23.26	24.56	25.71	26.34	27.67	28.34

Table 6
Comparison of latency.

Algorithm	100	200	300	400	500	600
<i>FAST</i>	425	435	439	443	452	461
<i>MSS</i>	413	419	423	428	431	435
<i>DMPT</i>	423	427	431	435	439	441
<i>AEM-Tree</i>	452	459	462	474	481	494
<i>EMT</i>	386	391	399	401	415	423

slightly less steep growth, with values rising from 357 for 100 transactions to 462 for 600 transactions. In contrast, the *DMPT* algorithm maintains relatively stable *TRT* values across the range of Transaction Counts, with only moderate increases from 351 to 402. The *AEM-Tree* and *EMT* algorithms also show increasing *TRT* with higher Transaction Counts, but their growth rates are relatively modest, starting at 341 and 312, respectively, and ending at 379 and 374 for 600 transactions. The study emphasizes the significance of selecting an algorithm that matches the specific scalability and response time requirements of an application's workload. Table 3 provides a comprehensive comparative analysis of TVT across various algorithms at different input sizes. Notably, the *FAST*, *MSS*, *DMPT*, and *AEM-Tree* algorithms exhibit a consistent linear increase in TVT as the input size grows. While the growth rates may vary slightly, this predictability suggests that these algorithms can effectively handle larger input sizes while maintaining relatively stable TVT. In contrast, the *EMT* algorithm distinguishes itself by reliability, delivering notably lower TVT values than the others, indicating faster transaction verification across the entire spectrum of input sizes. This exceptional efficiency could make *EMTs* compelling for applications requiring swift transaction verification. However, selecting the most suitable algorithm should be guided by carefully considering the specific requirements and trade-offs within the context of the intended blockchain application. Table 5 provides a comparative analysis of Latency, a critical metric for assessing the responsiveness and efficiency of many algorithms at different input sizes. Notably, the *reliability of the FAST algorithm* delivers low latency values across all input sizes, ranging from 425 m to 461 m. Similarly, the *MSS* algorithm maintains low latency, with 413 m and 435 m values. The *DMPT* algorithm exhibits latency within a narrow range, from 423 m to 441 m, displaying stable and predictable performance. However, the *AEM-Tree* algorithm records slightly higher latency values, ranging from 452 m to 494 m., indicating a somewhat less efficient performance than *FAST*, *MSS*, and *DMPT*.

In contrast, the *EMT* algorithm exhibits the highest latency values among the algorithms, with values between 386 m and 423 m. While efficient, latency remains considerably higher than the other algorithms, suggesting a potential trade-off between efficiency and responsiveness. A blockchain application's optimal algorithm selection should consider latency requirements to balance efficiency and availability.

5. Conclusion and future work

According to research, blockchain technology's fundamental features, such as permanence and cryptographic methods, are capable of helping secure sensitive data within the Big Data (BD) setting, thus improving data security and privacy. This research shows that mathematically adjusted Merkle Trees (MT) significantly improve data

integrity and verification compared to conventional MTs. Additionally, MTs incorporating cryptographic Hash Functions (HF) and mathematical algorithms further improve accuracy. The modification maintains the blockchain's integrity and enables quick identification of changes in data stored in blocks. The mathematically adjusted MT enhances privacy in the blockchain, and innovative encryption methods add another layer of confidentiality. The processing of BD relies heavily on modifications to guarantee private transactions, secure data access, and improve privacy. Scaling and optimizing the mathematically adjusted MT is necessary for future studies. Data security and confidentiality can be further enhanced by investigating novel mathematical notions and forming new cryptographic functions.

Furthermore, assessing the efficiency and performance of this updated structure in actual blockchain applications across several industry sectors would offer insightful information for helpful deployment.

Data availability

Not Applicable.

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CRediT authorship contribution statement

Yu Wang: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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